

Individual Project Work Plan

An Autonomous Robot for Indoor Navigation Using A* Search

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Course	CSE440	Section	2

Project Overview

This work plan will support the development of a software-based simulation of an autonomous robot for indoor navigation. No physical robot or hardware component will be used. The main focus will be the creation and visualization of a two-dimensional occupancy grid containing free cells and blocked cells. The software will allow maps to be created or loaded, obstacles to be placed or removed, and the starting point, destination, explored cells, calculated path, and virtual robot to be displayed clearly. The map will be connected with the A* path-planning and robot-simulation modules. A simple Python graphical interface will provide controls for map preparation, simulation operation, and dynamic obstacle updates.

Weekly Work Summary

Week	Date	Important Planned Work
1	11-17 July	Mapping requirements, GUI structure, software workflow, and sample grid layouts will be prepared.
2	18-24 July	The occupancy grid, map creation and loading, and obstacle placement system will be developed.
3	25-31 July	Visual map states will be added and the map will be connected with the A* module.
4	1-7 August	The robot position, calculated path, travelled cells, and remaining route will be displayed.
5	8-14 August	Dynamic obstacle placement and live map updating will be added.
6	15-21 August	The graphical interface and simulation controls will be completed and improved.
7	22-28 August	Map and GUI testing, code organization, screenshots, and documentation will be completed.

Week 1: 11 July - 17 July

I will define the software requirements for environment mapping, map loading, obstacle placement, and visual presentation. I will prepare the Python development environment and organize separate modules for the map and graphical interface. I will design the workflow for creating or loading a map, selecting the start and destination, placing obstacles, and displaying the environment. I will also prepare several sample grid layouts for later development and testing.

Week 2: 18 July - 24 July

I will develop the two-dimensional occupancy grid. Free cells will be represented by 0, while blocked cells will be represented by 1. I will add functions for creating new maps, loading saved maps, and placing or removing obstacles. Different grid sizes and obstacle arrangements will be tested to confirm that the map data is stored and updated correctly.

Week 3: 25 July - 31 July

I will add separate visual states for the starting point, destination, obstacles, explored cells, calculated path, and virtual robot. The map module will be connected with the A* path-planning module so that the selected grid, start position, and destination can be provided as input. I will verify that blocked cells and valid movement areas are transferred correctly and that the returned path can be displayed on the grid.

Week 4: 1 August - 7 August

I will display the virtual robot position on the grid after the A* module generates a valid path. The calculated path, travelled cells, and remaining route will be shown during navigation. The map display will be refreshed after every movement so that the robot can be observed clearly. I will test the visualization using short paths, long paths, narrow passages, and maps containing many obstacles.

Week 5: 8 August - 14 August

I will add an option for introducing a new obstacle while the robot is moving. The selected cell will be marked as blocked and the live map will be updated immediately. The changed map will be provided to the path-planning and simulation modules when the new obstacle affects the current route. Different obstacle positions will be tested to confirm that map updates are accurate and visible.

Week 6: 15 August - 21 August

I will improve the graphical user interface of the simulation. Clear visual indicators will distinguish free cells, obstacles, the starting point, destination, explored cells, final path, and virtual robot. Controls will be added for creating or loading a map, choosing the start and destination, placing obstacles, running or pausing the simulation, clearing the grid, and resetting the system. The layout and control behavior will be tested to make the interface simple and consistent.

Week 7: 22 August - 28 August

I will test the map and graphical interface using small and large grids, simple and complex obstacle patterns, blocked destinations, and dynamic obstacles. I will verify map loading, obstacle placement and removal, visual state updates, path display, robot-position display, and all interface controls. I will organize the source code, add comments, and prepare screenshots, test results, and project documentation for the final demonstration.

Expected Outcome

By the end of the seventh week, the software will allow a user to create or load an indoor occupancy grid, place or remove obstacles, select a starting point and destination, and view the explored cells, calculated path, and virtual robot. The live map will update when a dynamic obstacle is introduced, and the graphical interface will provide clear controls for preparing and operating the simulation. The completed work will improve my understanding of grid-based mapping, Python programming, graphical interface development, modular software integration, and systematic testing.

Conclusion

This seven-week plan will guide the development of the environment-mapping and graphical-interface components of the software-only indoor navigation simulator. The work will progress from requirements and occupancy-grid development to module integration, live map updates, interface improvement, and final testing. All weekly activities are planned future tasks and will be completed according to the stated schedule.